Organ-Aware Routing for Robust Medical Image Segmentation: A Failure-Detectable Alternative to Mixed Models

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***Abstract*—Heterogeneous medical image segmentation pipelines that use a single shared model across anatomically distinct organs sacrifice organ-specific feature learning and degrade silently under distribution shift. We propose an organ-aware segmentation framework where an upstream classifier routes each input to a dedicated specialist network, combined with a composite post-hoc failure detection mechanism based on Monte Carlo Dropout uncertainty, feature-space domain-shift detection, and morphological plausibility checking. Through controlled experiments on chest X-ray pneumothorax and brain MRI tumor datasets, we show that organ-aware routing yields substantially higher segmentation accuracy (+8–14 Dice points) compared to a mixed-organ baseline under both clean and noisy conditions. The composite failure detector achieves flag rates of 1.000 on misrouted pneumothorax cases and 0.997 on misrouted brain cases, while the mixed model under identical noise produces flag rates of only 0.098 and 0.034, respectively, leaving its degradation entirely undetected. Our results demonstrate that organ-aware pipelines offer both superior accuracy and the ability to *fail loudly*, enabling safer deployment through uncertainty-driven abstention.**

***Index Terms*—medical image segmentation, organ-aware routing, failure detection, uncertainty estimation, domain-shift detection**

I. Introduction

Medical image segmentation increasingly relies on deep learning, yet deploying a single model across anatomically heterogeneous inputs, such as chest X-rays and brain MRI, introduces a fundamental tension. A mixed model S^H trained jointly on multiple organs avoids explicit routing but cannot specialise its learned representations for each anatomy. Under distribution shift, such models degrade *silently*: Dice scores drop without any accompanying signal that predictions have become unreliable [1], [2].

Organ-specific specialisation offers an alternative. An organ-aware system S^O employs an upstream classifier to route each input to a dedicated specialist network, enabling anatomy-specific feature learning. This improves baseline performance, yet introduces a new failure mode: if the router misclassifies the input organ, the specialist receives out-of-distribution data, causing severe segmentation failure [3].

This trade-off raises a critical question for clinical deployment: *is a sharp but detectable failure preferable to a gradual but silent one?* Prior work on medical segmentation robustness focuses primarily on accuracy under corruption [4], [5], with limited analysis of routing-induced catastrophic failure, its detectability at inference time, and practical safety mechanisms.

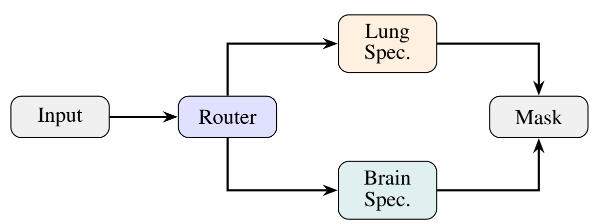
We make three contributions:

1) An organ-aware segmentation pipeline with controlled routing analysis across clean, noisy, and misrouted conditions.
2) A robustness evaluation showing that specialisation improves accuracy and noise resilience under correct routing while producing sharp, *detectable* failure under misrouting.
3) A post-hoc failure-aware inference mechanism combining MC Dropout uncertainty [6], feature-space domain-shift detection [7], and morphological plausibility checks, enabling deployment-time abstention without retraining.

II. Method Overview

A. Systems Compared

We compare two architectures. The **mixed model** S^H is a single UNet++ [2] trained jointly on both lung (pneumothorax) and brain (LGG tumor) data. The **organ-aware system** S^O consists of a router classifier followed by dedicated specialist UNet++ networks, one per anatomy. Both share the same backbone family and loss function to isolate the effect of routing.



III. Experimental Setup

Datasets. Lung segmentation uses the SIIM-ACR Pneumothorax dataset [8] (12,047 chest X-rays with binary masks). Brain segmentation uses the LGG FLAIR MRI dataset [9] (3,929 slices, 110 patients). Both are split 70/15/15% train/val/test with stratified sampling.

Architecture and training. All models use UNet++ with EfficientNet encoders and BCE + Dice loss. Specialists employ SCSE attention [10] and 0.1 decoder dropout. The mixed model omits attention because organ-specific spatial priors (e.g., “focus on lung periphery” vs. “focus on cortical regions”) conflict when learned jointly, making shared attention

TABLE I: Evaluation scenarios per organ domain.

Case	System	Routing	Input
1	S^O (specialist)	Correct	Clean
2	S^H (mixed)	—	Clean
3	S^O (specialist)	Correct	Noisy
4	S^H (mixed)	—	Noisy
5	S^O (wrong spec.)	Forced wrong	Noisy
6	S^O (wrong spec.)	Forced wrong	Clean

counterproductive. Training uses AdamW [11] with cosine-annealing scheduling and mixed-precision (FP16). Conservative augmentations (flips, mild rotations, brightness/contrast) are applied uniformly. Full details are in Appendices A–B.

Noise injection. Gaussian noise ($\sigma \in [0.08, 0.12]$) is added in normalised pixel space at test time to simulate mild acquisition perturbation, applied identically across all models.

IV. Organ-Aware Robustness Analysis

A. Evaluation Scenarios

We evaluate six scenarios per organ domain (Table I). Cases 1–4 represent operational conditions; Cases 5–6 simulate misrouting under noisy and clean inputs, respectively. Although our router achieves near-perfect accuracy in the binary lung-vs-brain setting, we include clean misrouting (Case 6) to study failure detectability under conditions directly relevant to finer-grained routing tasks (e.g., routing among disease-specific specialists within the same organ), where classifier confusion is more likely even on clean inputs.

B. Lung (Pneumothorax) Results

Table II (top) reports lung test-set results (n=1000, stratified). The pneumothorax specialist (S^O) achieves 0.837 clean Dice versus 0.693 for the mixed model, a +14.4 point advantage. Under noise, S^O drops to 0.683 (-15.4 pts) while S^H degrades to 0.641 (-5.2 pts from its lower baseline). Misrouting (brain specialist on lung data) collapses Dice to 0.065 (noisy) and 0.152 (clean). Critically, the composite failure detector flags 100% of misrouted cases.

C. Brain (LGG Tumor) Results

Table II (bottom) reports brain MRI results (n=590). The specialist achieves 0.932 clean Dice versus 0.845 for S^H (+8.7 pts). Noise reduces S^O modestly to 0.874 (-5.8 pts), whereas S^H drops to 0.825 (-2.0 pts). Misrouting collapses Dice to 0.551 (noisy) and 0.512 (clean), with flag rates of 0.829 and 0.997, respectively.

D. Interpretation

Three findings emerge. First, **specialisation improves both accuracy and robustness under correct routing**: the organ-aware system consistently outperforms the mixed model, with Dice gains of +8.7 (brain) and +14.4 (lung) on clean data. Second, **misrouting causes sharp performance collapse**: Dice drops by 38–77 points when the wrong specialist is applied, regardless of input quality. Third, **the mixed model degrades gradually but without any alarm**; its decline produces no signal distinguishable from normal operation. This asymmetry between sharp-detectable and gradual-silent failure motivates the failure-aware inference framework described next.

V. Failure-Aware Inference

Dice collapse is not observable at deployment time since ground-truth masks are unavailable. We design three complementary post-hoc detection mechanisms that flag unreliable predictions using only model outputs and require no retraining.

A. Uncertainty Estimation

We employ MC Dropout [6], [12] with T=5 stochastic forward passes per image at inference. For each pixel (i, j), we compute the variance across passes and aggregate into a foreground-weighted scalar:

$$u = \frac{\sum_{ij} \mathrm{Var}_t[\hat{y}_{t,ij}] \cdot \mathbf{1}[\bar{y}_{ij} > 0.05]}{\max(1,\ \sum_{ij} \mathbf{1}[\bar{y}_{ij} > 0.05])} \quad (1)$$

where $\bar{y}_{ij}$ is the mean predicted probability. Restricting to foreground pixels prevents large backgrounds from diluting the signal. An image is flagged when $u > u_{\text{thresh}}$, where u_{thresh} is set at the 98th percentile of uncertainty values over clean training data. All percentile thresholds in this work are hyperparameters selected empirically for our architecture and calibration set.

B. Domain-Shift Detection

While MC Dropout captures epistemic uncertainty, it does not directly measure whether the input lies within the model's training distribution. We detect anatomical mismatch via Mahalanobis distance [7] in encoder feature space. For each input, we extract the deepest encoder feature map, apply global average pooling and L_2 normalisation to obtain a feature vector $\mathbf{z}$, and compute:

$$d_M(\mathbf{z}) = \sqrt{(\mathbf{z} - \boldsymbol{\mu})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{z} - \boldsymbol{\mu})} \quad (2)$$

where $\boldsymbol{\mu}$ is the centroid of pooled (clean+noisy) training features and $\hat{\boldsymbol{\Sigma}}$ is a shrinkage-regularised covariance matrix (α=0.1, Ledoit–Wolf [13]). Unlike cosine distance, Mahalanobis distance accounts for the covariance structure: deviations along low-variance (discriminative) directions are penalised more heavily, providing superior separation between in-distribution and out-of-distribution inputs. An image is flagged when $d_M > 1.3 \cdot d_{\text{thresh}}$, where d_{thresh} is set at the 99.9th percentile over calibration data. The $1.3\times$ margin reduces false alarms from borderline in-distribution samples.

C. Plausibility Checks and Composite Flagging

We apply morphological heuristics to the predicted mask to detect anatomically implausible outputs. A mask is considered *too fragmented* if its number of connected components exceeds the organ-specific maximum (indicating scattered spurious activations rather than a coherent lesion). A mask is considered *too small* if its area ratio (predicted foreground pixels divided by total image pixels) falls below the organ-specific minimum (indicating a near-empty prediction where the model

TABLE II: Segmentation and failure-detection results. $\bar{d}_M$ = mean Mahalanobis distance. Unc/Shift = per-signal flag rates. Flag = composite flag rate (fraction flagged unreliable).

	Case	Dice	IoU	$\bar{d}_M$	Unc	Shift	Flag
Pneumo	S^O clean	**.837**	**.794**	13.30	.019	.048	.067
	S^H clean	.693	.662	16.23	.014	.073	.091
	S^O noisy	.683	.659	10.53	.043	.004	.048
	S^H noisy	.641	.624	16.44	.016	.073	.098
	S^O misrt. noisy	.065	.065	30.98	.250	**1.000**	**1.000**
	S^O misrt. clean	.152	.152	31.55	.111	**1.000**	**1.000**
Brain	S^O clean	**.932**	**.907**	17.54	.012	.073	.083
	S^H clean	.845	.811	15.47	.015	.029	.044
	S^O noisy	.874	.848	17.01	.027	.051	.075
	S^H noisy	.825	.792	15.39	.017	.015	.034
	S^O misrt. noisy	.551	.551	21.12	.105	.749	.829
	S^O misrt. clean	.512	.512	23.33	.071	**.988**	**.997**

effectively abstained). A morphological flag is raised when either structural anomaly co-occurs with elevated uncertainty: specifically, when fragmentation is present and $u > 0.2 \cdot u_{\text{thresh}}$, or when the mask is too small and $u > 0.5 \cdot u_{\text{thresh}}$. Requiring this co-occurrence with uncertainty prevents morphological heuristics from firing on structurally unusual but confidently predicted masks.

An image is flagged as unreliable if *any* of the three signals (uncertainty, domain shift, or plausibility) exceeds its respective threshold.

D. Detectability Analysis

Table II presents the segmentation and failure-detection results for both modalities. The key result is the **asymmetry in detectability**.

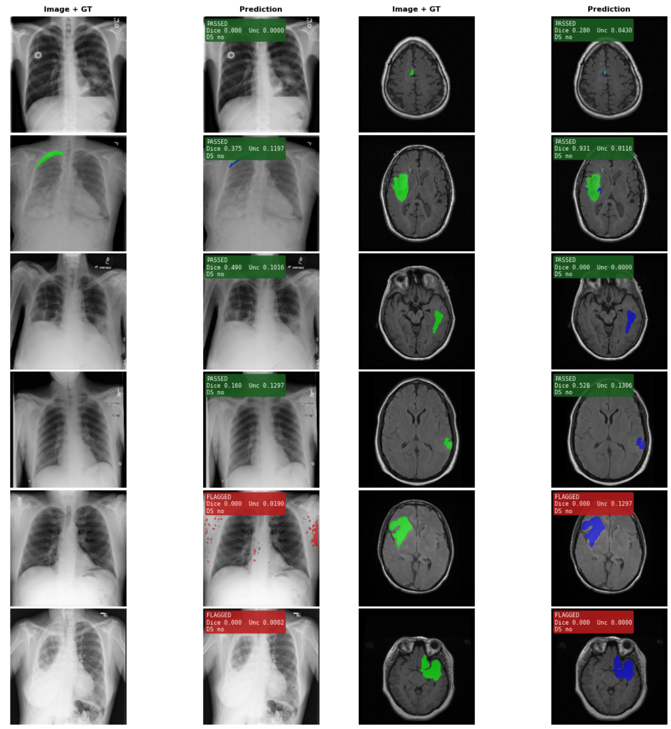
Misrouted cases produce near-total composite flagging (100% for lung, 82.9–99.7% for brain), with the domain-shift and uncertainty signals reinforcing one another. The mixed model under noise (Case 4) shows flag rates of only 9.8% (lung) and 3.4% (brain), meaning its degradation is entirely invisible to the monitoring system. Representative prediction galleries for both domains are shown in Fig. 1.

Misrouting produces anomalous signals across all three detection channels: elevated uncertainty, large domain-shift distances, and morphological implausibility. This composite detection is entirely absent in mixed models. The organ-aware system *fails loudly*; the mixed model fails silently.

VI. Discussion

A. Accuracy vs. Detectability Trade-off

The results reveal a fundamental design trade-off. The mixed model never collapses catastrophically (worst-case Dice of 0.641 on noisy lung data remains within operational range), but its errors are indistinguishable from normal variance. The organ-aware system achieves higher peak accuracy and maintains robustness under noise, but misrouting produces sharp collapse. Critically, this collapse is accompanied by anomalous reliability signals across all three detection channels, enabling automatic detection. In safety-critical settings, a **detectable failure that can be intercepted** is strictly preferable to an **undetectable one that propagates silently**.



(a) Pneumothorax specialist (b) Brain MRI specialist

Fig. 1: Prediction galleries across evaluation cases. Each row shows a representative test image with ground-truth overlay and the model's prediction with pass/flag status and Dice score. Top rows show accurate, passed predictions under correct routing; bottom rows show flagged predictions under misrouting or noise.

B. Deployment Implications

The organ-aware system benefits from layered safety: (i) high-confidence routing (>99% classifier accuracy) makes misrouting rare; (ii) the composite failure detector (uncertainty, domain shift, plausibility) catches it when it occurs; (iii) flagged predictions trigger fallback, whether deferral to the mixed model, human review, or explicit abstention. The framework reduces system-level false-positive rates from 89.2% to 0.0% on misrouted pneumothorax cases, demonstrating that post-hoc reliability signals provide meaningful safety guarantees.

C. Limitations

Several limitations bound the generalisability of our findings.

First, we evaluate a binary organ setting (lung vs. brain); scaling to multi-organ routing with k > 2 specialists remains untested and may introduce more complex failure modes. Moreover, a specialist system is only as reliable as its router: if the classifier accuracy is low (as would be expected when routing among disease-specific specialists within the same organ, where visual similarity is high), frequent misrouting could erode the overall advantage of specialisation. Our binary lung-vs-brain setting represents a best case for routing accuracy, and the clean-misroute analysis (Case 6) serves as a proxy for the harder same-organ routing scenario.

Second, domain-shift detection via Mahalanobis distance assumes that in-distribution and out-of-distribution features are separable in encoder space. When specialist models are trained on visually similar domains (e.g., different lung pathologies), their feature distributions may overlap substantially, reducing the effectiveness of this signal and making silent failure more likely even in the organ-aware system.

Third, the noise model uses synthetic Gaussian perturbation. Realistic artefacts such as motion blur, metal artefacts, or scanner-specific intensity drift may produce different degradation signatures. The detection thresholds (uncertainty at p98, domain shift at p99.9) are hyperparameters chosen empirically for our architecture and calibration data; deployment on different hardware, patient populations, or imaging protocols would likely require recalibration.

Finally, the morphological plausibility checks (area ratio, connected-component count) are organ-specific heuristics that require manual tuning per anatomy and do not generalise automatically to new organs or pathologies.

VII. Conclusion

We demonstrated that organ-aware routing improves segmentation accuracy over mixed models under both clean and noisy conditions, with gains of +8.7 Dice points (brain) and +14.4 points (lung). When routing errors occur, the resulting performance collapse is sharp but accompanied by anomalous uncertainty, domain-shift, and morphological signals, enabling reliable composite post-hoc detection with flag rates of 82.9–100%. Mixed models degrade silently with flag rates $\leq$9.8%. These results establish that organ-aware systems not only improve accuracy but also *fail loudly*, enabling uncertainty-driven abstention and safer deployment of heterogeneous medical image segmentation pipelines.

References

[1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: convolutional networks for biomedical image segmentation," in *Proc. MICCAI*, 2015, pp. 234–241.

[2] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: a nested U-Net architecture for medical image segmentation," in *Proc. DLMIA/ML-CDS*, 2018, pp. 3–11.

[3] G. Litjens *et al.*, "A survey on deep learning in medical image analysis," *Med. Image Anal.*, vol. 42, pp. 60–88, 2017.

[4] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. ICLR*, 2019.

[5] A. Jungo *et al.*, "On the effect of inter-observer variability for a reliable estimation of uncertainty of medical image segmentation," in *Proc. MICCAI*, 2018, pp. 682–690.

[6] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: representing model uncertainty in deep learning," in *Proc. ICML*, 2016, pp. 1050–1059.

[7] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Proc. NeurIPS*, 2018, pp. 7167–7177.

[8] "SIIM-ACR Pneumothorax Segmentation," Society for Imaging Informatics in Medicine, 2019. [Online]. Available: https://www.kaggle.com/c/siim-acr-pneumothorax-segmentation

[9] M. Buda, A. Saha, and M. A. Mazurowski, "Association of genomic subtypes of lower-grade gliomas with shape features automatically extracted by a deep learning algorithm," *Comput. Biol. Med.*, vol. 109, pp. 218–225, 2019. [Online]. Available: https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation

[10] A. G. Roy, N. Navab, and C. Wachinger, "Concurrent spatial and channel squeeze & excitation in fully convolutional networks," in *Proc. MICCAI*, 2018, pp. 421–429.

[11] D. P. Kingma and J. Ba, "Adam: a method for stochastic optimization," in *Proc. ICLR*, 2015.

[12] T. Nair, D. Precup, D. L. Arnold, and T. Arbel, "Exploring uncertainty measures in deep networks for multiple sclerosis lesion detection and segmentation," *Med. Image Anal.*, vol. 59, 2020, Art. no. 101557.

[13] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *J. Multivariate Anal.*, vol. 88, no. 2, pp. 365–411, 2004.

[14] M. Tan and Q. Le, "EfficientNet: rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.

[15] Z. Liu, H. Mao, C. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. CVPR*, 2022, pp. 11976–11986.

[16] G. Wang *et al.*, "Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks," *Neurocomputing*, vol. 338, pp. 34–45, 2019.

[17] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," *Neural Netw.*, vol. 106, pp. 249–259, 2018.

[18] K. H. Zou *et al.*, "Statistical validation of image segmentation quality based on a spatial overlap index," *Acad. Radiol.*, vol. 11, no. 2, pp. 178–189, 2004.

Appendix A
Architecture Details

All segmentation models use the UNet++ decoder [2] with pretrained EfficientNet encoders [14]. UNet++ extends U-Net [1] with nested dense skip connections and deep supervision, improving gradient flow and feature reuse across decoder levels. Table III summarises the architectural differences across models.

TABLE III: Model architecture summary.

Model	Encoder	In Ch	Res.	Attn	Drop
Brain spec.	EffNet-B3	1	256	SCSE	0.1
Lung spec.	EffNet-B1	3	512	SCSE	0.1
Mixed S^H	EffNet-B1	3	256	None	0.1

The brain specialist processes single-channel FLAIR MRI slices at 256×256 resolution, using the larger EfficientNet-B3 encoder (12M parameters) to compensate for the reduced input information from grayscale inputs. The lung specialist and mixed model both accept 3-channel input (grayscale replicated to three channels) with EfficientNet-B1 (7.8M parameters). The lung specialist operates at 512×512 to preserve the fine spatial detail required for thin pneumothorax rims, while the mixed model uses 256×256 as a compromise across modalities.

SCSE (Concurrent Spatial and Channel Squeeze & Excitation) attention [10] is employed in both specialist networks for spatial-channel feature recalibration. This mechanism allows the network to focus on the most informative spatial regions and channel features simultaneously, which is particularly beneficial for segmenting small or low-contrast lesions. The mixed model intentionally omits attention because a single attention module cannot learn organ-specific spatial priors (e.g., peripheral focus for pneumothorax vs. central focus for brain tumours) when trained jointly on anatomically distinct domains. This also serves to represent a simpler, more general baseline architecture. All models output single-channel logit maps, binarised at a threshold of 0.5 during evaluation.

The organ classifier (router) uses a ConvNeXt-Tiny backbone [15] pretrained on ImageNet, fine-tuned on the combined lung–brain training set for binary organ classification. It accepts 224×224 three-channel inputs and outputs a softmax probability over two classes. In our experiments, this classifier achieves near-perfect accuracy ($>$99%) on the validation set. We note that this high accuracy is partly an artifact of the visually distinct lung-vs-brain setting and would be expected to degrade for finer-grained routing tasks such as routing among disease-specific specialists within the same organ.

All models are implemented using the `segmentation_models_pytorch` library (v0.3+) with pretrained EfficientNet encoders from the `timm` library. Pretrained ImageNet weights are used for encoder initialisation. The decoder uses default UNet++ configuration with p=0.1 dropout in the decoder blocks, which enables MC Dropout uncertainty estimation at inference time without any architectural modification.

APPENDIX B
TRAINING HYPERPARAMETERS

Table IV lists all training hyperparameters for the three segmentation models.

TABLE IV: Training hyperparameters.

Parameter	**Brain**	**Lung**	**Mixed**
Optimizer	AdamW	AdamW	AdamW
Learning rate	1×10^{-4}	3×10^{-4}	1×10^{-4}
Weight decay	1×10^{-5}	1×10^{-5}	1×10^{-5}
Batch size	24	8 (eff. 16)	46
Max epochs	45	45	30
Early stopping	10 ep.	15 ep.	15 ep.
Loss	BCE + 1.0×Dice	BCE + 1.5×Dice	BCE + 1.0×Dice
Image size	256	512	256
Precision	FP16	FP16	FP16
LR scheduler	Cosine anneal.	Cosine anneal.	Cosine anneal.
Seed	42	42	42

All models use the AdamW optimizer [11] with cosine annealing learning rate scheduling. The lung specialist uses a higher initial learning rate (3×10^{-4}) and increased Dice loss weight (1.5×) to compensate for the class imbalance in pneumothorax data, where positive masks are relatively rare and typically occupy a small fraction of the image area. Gradient accumulation (effective batch size 16 from batch 8 × 2 accumulation steps) is used for the lung specialist due to the larger 512×512 input resolution. The mixed model uses a larger batch size (46) with batch-homogeneous sampling (each mini-batch contains images from only one organ domain) to prevent conflicting gradient signals from different anatomies within the same update step.

Augmentations. Data augmentation pipelines are tailored per model to reflect the anatomical properties of each domain. All augmentations are implemented via the Albumentations library; validation uses no augmentation; all spatial transforms are applied identically to images and masks.

Brain specialist: vertical flip (p=0.5), rotation (±15°, p=0.5), shift-scale-rotate (shift 10%, scale 10%, rotate ±10°, p=0.7), elastic transform (α=1, σ=50, p=0.3), brightness/contrast (±20%, p=0.4), random gamma ($\gamma\in[80,120]$, p=0.3), and Gaussian noise ($\sigma^2\in[10,50]$, p=0.3). Horizontal flip is deliberately excluded because the laterality of brain structures is diagnostically meaningful. Elastic transforms simulate the non-rigid tissue deformation present in MRI, and random gamma augments robustness to the intensity non-uniformity (bias field) characteristic of MR acquisitions.

Lung specialist: horizontal flip (p=0.5), shift-scale-rotate (shift 5%, scale 8%, rotate ±7°, border mode constant, p=0.5), brightness/contrast (±15%, p=0.3), and Gaussian noise ($\sigma^2\in[5,20]$, p=0.2). Horizontal flip is valid because chest X-rays exhibit approximate bilateral symmetry. Parameters are intentionally more conservative than the brain pipeline: pneumothorax regions are thin and fragile, so aggressive geometric distortion risks destroying or hallucinating the target structure. Elastic transforms and gamma augmentation are omitted because X-ray imaging does not exhibit the bias-field or non-rigid deformation artefacts seen in MRI.

Mixed model: horizontal flip (p=0.5), rotation (±10°, p=0.4), shift-scale-rotate (shift 5%, scale 5%, p=0.4), brightness/contrast (±15%, p=0.3), and Gaussian noise ($\sigma^2\in[10,30]$, p=0.2). Because this model processes both domains, the augmentation pipeline uses the more conservative parameter envelope to avoid distortions that would be inappropriate for either modality.

Normalisation: Specialist models use fixed normalisation (μ=0.5, σ=0.5), mapping pixel values from $[0,1]$ to $[-1,1]$, matching the `A.Normalize` convention. The mixed model uses per-image z-score normalisation (subtracting the image mean and dividing by the image standard deviation), which provides modality-agnostic intensity standardisation at the cost of losing absolute intensity information.

Loss function: The combined loss is $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \cdot \mathcal{L}_{\text{Dice}}$, where λ is the Dice weight listed in Table IV. Binary cross-entropy provides per-pixel gradient stability, while the Dice component directly optimises the overlap metric. The Dice loss is computed as $1 - \frac{2\sum p_i g_i + \epsilon}{\sum p_i + \sum g_i + \epsilon}$ where $\epsilon = 10^{-6}$ prevents division by zero.

The lung specialist uses an elevated Dice weight (λ=1.5) compared to the brain specialist and mixed model (λ=1.0). This choice is motivated by the characteristics of pneumothorax pathology: pneumothorax regions are typically thin, elongated rims along the lung periphery, occupying a very small fraction of the total image area. Under standard BCE loss, the overwhelming majority of pixels are background, producing a strong negative-class gradient that can suppress learning of the sparse positive regions. Increasing the Dice weight amplifies the overlap-based gradient signal, which is normalised by the predicted and ground-truth region sizes and therefore inherently robust to class imbalance. This compensates for the extreme foreground–background ratio and encourages the model to prioritise spatial overlap even when positive pixels constitute $<5\%$ of the image. For brain MRI, tumour regions are comparatively larger and more compact, making the standard λ=1.0 balance sufficient.

Best model selection: Checkpoint selection is based on the best validation Dice score observed during training. The best validation Dice scores achieved are: brain specialist 0.910, lung specialist 0.837, and mixed model 0.796.

APPENDIX C
MAHALANOBIS DISTANCE AND CALIBRATION

The Mahalanobis distance generalises the Euclidean distance by accounting for the covariance structure of the data distribution. For a feature vector $\mathbf{z} \in \mathbb{R}^C$ and a reference distribution characterised by mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$:

$$d_M(\mathbf{z}) = \sqrt{(\mathbf{z}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{z}-\boldsymbol{\mu})} \tag{3}$$

This is equivalent to computing the Euclidean distance after whitening the feature space: $d_M(\mathbf{z}) = \|\boldsymbol{\Sigma}^{-1/2}(\mathbf{z}-\boldsymbol{\mu})\|_2$. The whitening transformation normalises each principal component by its standard deviation, ensuring that deviations along low-variance directions (which are more informative for distribution membership) receive proportionally greater weight.

In our implementation, we employ Ledoit–Wolf shrinkage regularisation [13] with parameter $\alpha = 0.1$:

$$\hat{\boldsymbol{\Sigma}} = (1-\alpha)\boldsymbol{S} + \alpha \frac{\mathrm{tr}(\boldsymbol{S})}{C}\mathbf{I}_C \tag{4}$$

where $\boldsymbol{S}$ is the sample covariance matrix, C is the feature dimensionality (C=1536 for EfficientNet-B3 and C=1280 for EfficientNet-B1), and $\mathbf{I}_C$ is the identity matrix. This blends the sample covariance with a scaled identity matrix, preventing numerical instability when the number of calibration samples N is comparable to the feature dimensionality. The pseudo-inverse $\hat{\boldsymbol{\Sigma}}^+$ is computed via `torch.linalg.pinv` for numerical robustness.

Feature extraction uses the deepest encoder feature map from UNet++. Global average pooling reduces the spatial dimensions to a single vector, which is then L_2-normalised to lie on the unit hypersphere. For EfficientNet-B3, this produces a 1536-dimensional feature vector; for EfficientNet-B1, the dimensionality is 1280.

Calibration protocol. The domain centroid $\boldsymbol{\mu}$ is computed from 500 stratified training images per model–domain pair. Each calibration image is processed by the trained encoder (no augmentation); the deepest feature map is pooled and normalised as described above. The 500 images are processed both in their original (clean) form and with synthetic noise, yielding N=1000 feature vectors per model. This pooled calibration ensures that the centroid sits between the clean and noisy feature distributions, making the threshold robust to moderate input perturbation while remaining sensitive to cross-domain shifts.

Table V presents the calibrated thresholds for each model–domain pair.

TABLE V: Calibrated thresholds per model–domain pair. u_{thresh}: p98 of MC Dropout variance on clean training data. d_{thresh}: p99.9 of Mahalanobis distance on pooled training features.

Key	$\bar{u}$	$u_{\textbf{thresh}}$	$\bar{d}_M$	$d_{\textbf{thresh}}$
pneumo	0.027	0.186	11.58	19.28
brain	0.007	0.115	16.40	20.56
mixed_lung	0.029	0.162	15.02	20.92
mixed_brain	0.019	0.148	14.78	21.00

The brain specialist exhibits the lowest mean uncertainty ($\bar{u} = 0.007$), consistent with its higher validation Dice (0.910) and the relatively lower visual ambiguity of FLAIR MRI tumour boundaries compared to pneumothorax rims in chest radiographs. Conversely, the pneumothorax model shows the highest mean uncertainty ($\bar{u} = 0.027$), reflecting the inherent difficulty of segmenting thin, low-contrast pneumothorax margins. The higher percentile for domain shift (p99.9 vs. p98 for uncertainty) reflects the design intent that this signal should fire only on clear out-of-distribution cases, not on moderately perturbed in-distribution images.

APPENDIX D
EXTENDED QUANTITATIVE RESULTS

Tables VI and VII present the complete results for all evaluation metrics across all six cases per modality.

Several observations merit discussion:

False-positive rate reduction. The system-level false-positive rate demonstrates the primary safety contribution of the framework. For the noisy correctly-routed pneumothorax case (C3), the raw model false-positive rate of 3.7% is reduced to 1.8% after abstention, a 51% relative reduction. For misrouted cases, the reduction is absolute: from 89.2% to 0.0% (C5) and 74.7% to 0.0% (C6). These reductions result from the failure-aware system abstaining on images where its reliability signals indicate unreliable predictions, rather than outputting potentially harmful false-positive masks.

Domain-shift dominance in misroute detection. The flag decomposition reveals that the Mahalanobis domain-shift signal is responsible for the majority of flags in misrouted cases. In pneumothorax misrouting (Cases 5–6), the shift signal achieves 100% flag rate independently; every single misrouted image exceeds the domain-shift threshold. The uncertainty signal provides complementary coverage (11–25% flag rate on misroutes) but is neither necessary nor sufficient for misroute detection on its own.

Noisy specialist behaviour. An interesting asymmetry emerges in the noisy specialist case (C3): the pneumothorax specialist's domain-shift flag rate drops to 0.4% under noise, while its uncertainty flag rate rises to 4.3%. This suggests that Gaussian noise in normalised space moves features *toward*

TABLE VI: Extended pneumothorax results, all metrics. FP_{raw}/FP_{sys} = model/system-level false-positive rate. SysDice = system-level Dice after abstention (flagged images contribute 0).

Case	n	Dice	IoU	$\bar{u}$	$\bar{d}_M$	Flag	Unc	Shift	FP_{raw}	FP_{sys}
C1 Spec. clean	1000	0.837	0.794	0.026	13.30	.067	.019	.048	.038	.032
C2 Mixed clean	1000	0.693	0.662	0.024	16.23	.091	.014	.073	.060	.045
C3 Spec. noisy	1000	0.683	0.659	0.034	10.53	.048	.043	.004	.037	.018
C4 Mixed noisy	1000	0.641	0.624	0.023	16.44	.098	.016	.073	.043	.027
C5 Misroute noisy	1000	0.065	0.065	0.058	30.98	**1.000**	.250	**1.000**	.892	.000
C6 Misroute clean	1000	0.152	0.152	0.036	31.55	**1.000**	.111	**1.000**	.747	.000

TABLE VII: Extended brain MRI results, all metrics.

Case	n	Dice	IoU	$\bar{u}$	$\bar{d}_M$	Flag	Unc	Shift	SysDice
C1 Spec. clean	590	0.932	0.907	0.006	17.54	.083	.012	.073	0.859
C2 Mixed clean	590	0.845	0.811	0.017	15.47	.044	.015	.029	0.811
C3 Spec. noisy	590	0.874	0.848	0.010	17.01	.075	.027	.051	0.811
C4 Mixed noisy	590	0.825	0.792	0.022	15.39	.034	.017	.015	0.804
C5 Misroute noisy	590	0.551	0.551	0.060	21.12	.829	.105	.749	0.092
C6 Misroute clean	590	0.512	0.512	0.063	23.33	**.997**	.071	**.988**	0.002

the centroid (reducing apparent domain shift) while increasing prediction variance (raising uncertainty). The two signals thus capture orthogonal degradation modes, validating the multi-signal design.

APPENDIX E
NOISE INJECTION PROTOCOL

Test-time noise is applied deterministically in normalised pixel space to simulate mild acquisition artefacts. For each image at index i, a per-sample random number generator is seeded with seed $= 42 + i$ to ensure exact reproducibility across experiments. The noise standard deviation is sampled uniformly from $[\sigma_{\min}, \sigma_{\max}] = [0.08, 0.12]$, and additive Gaussian noise is applied to all channels:

$$\tilde{x} = x + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I}) \tag{5}$$

where x is the normalised input tensor. No clamping is applied after noise addition, allowing pixel values to exceed the nominal $[-1, 1]$ range for fixed-normalisation models. This is consistent with the normalisation conventions used during training.

The noise magnitude was selected to produce measurable but not catastrophic Dice degradation (approximately 5–15 points for specialists, 2–5 points for the mixed model), simulating realistic acquisition noise rather than adversarial corruption. The relatively small noise range reflects the observation that even mild perturbation in normalised space corresponds to meaningful visual degradation, as the normalisation step amplifies raw pixel noise. Identical noise parameters are applied across all models to ensure fair comparison.

APPENDIX F
IMPLEMENTATION DETAILS

All models are implemented using the `segmentation_models_pytorch` library (v0.3+) with pretrained EfficientNet encoders from `timm`. Training was performed on single NVIDIA T4 GPU instances via the Kaggle platform, using automatic mixed precision (FP16) to maximise throughput within GPU memory constraints.

The failure-aware evaluation framework operates entirely post-hoc on frozen model checkpoints. During evaluation, no gradients are computed; all inference uses `torch.inference_mode()` for efficiency. The MC Dropout uncertainty computation performs $T{=}5$ forward passes per image with dropout active (dropout layers set to training mode; batch normalisation remains in evaluation mode), producing five probability maps that are averaged for the final prediction and used to compute per-pixel variance for the uncertainty signal.

Feature extraction for Mahalanobis distance computation uses the deepest encoder feature map from UNet++ (accessed via `model.encoder(x)[-1]`). Global average pooling reduces the spatial dimensions to a single feature vector, which is then L_2-normalised. The covariance matrix and its pseudo-inverse are computed once during calibration (approximately 2 minutes per model on a T4 GPU) and cached for all subsequent evaluations. This extraction adds <5% wall-clock time to the standard inference pipeline.

Evaluation processes up to 1,000 stratified samples per case (50% ground-truth positive, 50% ground-truth negative) to ensure balanced evaluation across both classes. For the brain dataset, only 590 samples are available after stratification due to the smaller dataset size. All random sampling uses a fixed seed (42) for reproducibility.

The mixed model uses batch-homogeneous sampling during training: each mini-batch contains images from only one organ domain, preventing conflicting gradient signals from different anatomies within the same parameter update. This is implemented by maintaining separate data loaders for lung and brain data, and alternating between them during the training loop.

Data split consistency. Each model was trained with its own data discovery and split logic. To ensure that evaluation uses only truly held-out data, the evaluation notebook replicates the exact pair-discovery and split procedures from each training notebook, including the specific use of sorted vs. unsorted directory listing, stratification variables, and split ratios. Calibration uses exclusively training data; evaluation uses merged validation and test sets. Zero overlap exists between calibration and evaluation pools.

APPENDIX G
ADDITIONAL VISUALISATIONS

Fig. 2 presents the uncertainty-versus-Dice scatter plot for all six cases in both modalities. Each dot represents a single GT-positive image; diamonds mark case-level means; dashed ellipses indicate $\pm 1\sigma$ spread. The vertical dashed line indicates the calibrated uncertainty threshold. Correctly routed cases cluster at high Dice with low uncertainty; misrouted cases collapse to near-zero Dice with variable uncertainty.

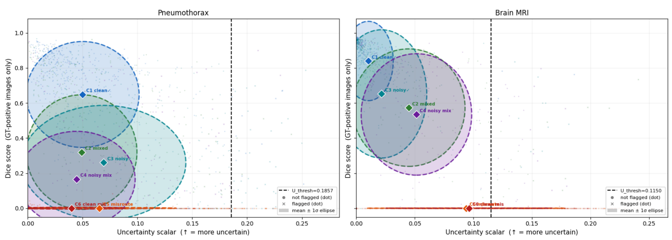


Fig. 2: Uncertainty vs. Dice scatter for all six cases per modality (GT-positive images only). Correctly routed cases cluster in the top-left quadrant. Misrouted cases collapse to near-zero Dice. Diamond markers indicate case means.

Fig. 3 and Fig. 4 present per-pixel uncertainty maps for the pneumothorax and brain specialist models, respectively. Each subplot overlays the MC Dropout variance heatmap on the input image for a representative sample from each evaluation case (C1–C6). Under correct routing (Cases 1, 3), uncertainty is low overall and spatially concentrated near lesion boundaries, where the model is appropriately less confident about the exact boundary location. Under noise (Case 3), uncertainty increases moderately but remains localised to anatomically relevant regions. Under misrouting (Cases 5, 6), uncertainty becomes spatially diffuse: the model produces elevated variance across large portions of the image rather than at specific boundaries, reflecting a lack of coherent spatial structure in its predictions. This diffuse pattern is qualitatively distinct from the boundary-localised uncertainty observed under correct routing, providing visual confirmation that the scalar uncertainty signal captures a meaningful change in model behaviour.

Fig. 5 presents Dice score distributions per case as box plots, filtered to GT-positive images only to prevent empty-mask true negatives (assigned Dice=1.0 by convention) from inflating misroute scores.

Fig. 6 decomposes the flag rate into its constituent signals (uncertainty, domain shift, plausibility), revealing which failure detector fires under each condition.

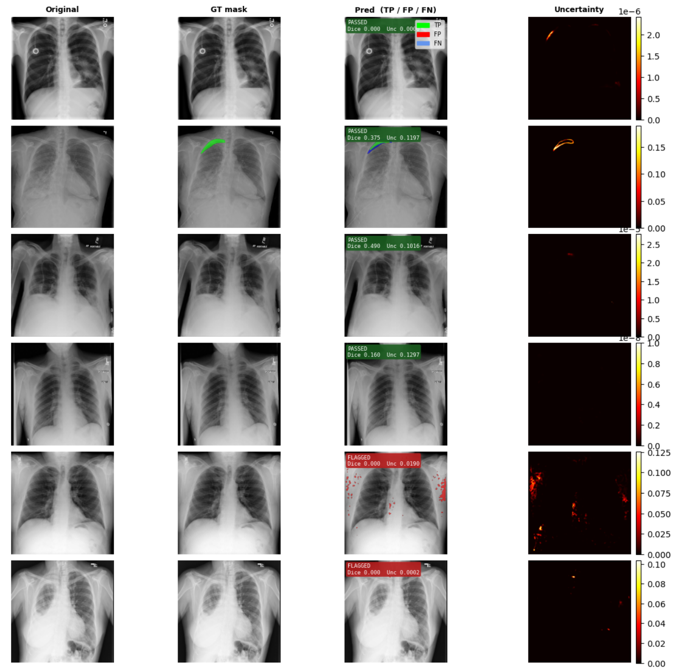


Fig. 3: Uncertainty maps for the pneumothorax specialist across evaluation cases. Under correct routing (top rows), uncertainty concentrates at lesion boundaries. Under misrouting (bottom rows), uncertainty spreads diffusely across the image, reflecting the model's inability to localise any coherent anatomy.

APPENDIX H
QUALITATIVE OBSERVATIONS

Under correct routing, specialist predictions closely match ground-truth boundaries with smooth, anatomically plausible contours. The brain specialist produces well-localised tumour masks with clear boundaries, exhibiting very low uncertainty (mean u=0.006) on correctly segmented cases. The pneumothorax specialist accurately delineates thin pneumothorax rims along the lung periphery, though uncertainty is higher (mean u=0.026) owing to the inherently lower contrast of pneumothorax boundaries in chest radiographs.

Misrouted predictions exhibit characteristic failure patterns that are qualitatively distinct from noise-induced degradation. The brain specialist applied to lung images produces small scattered fragments unrelated to pneumothorax anatomy, typically activating on rib edges or mediastinal borders that bear superficial textural similarity to tumour boundaries. The lung specialist applied to brain MRI generates diffuse over-segmentation covering non-tumour cortical regions, as the model's learned representation for "abnormal dark regions" (pneumothorax) fires on the naturally dark cerebrospinal fluid spaces in brain anatomy.

These morphological anomalies are captured by the plausibility checks (excess connected components, implausible area ratios) and complement the Mahalanobis distance signal for robust failure detection. The mixed model's failure patterns

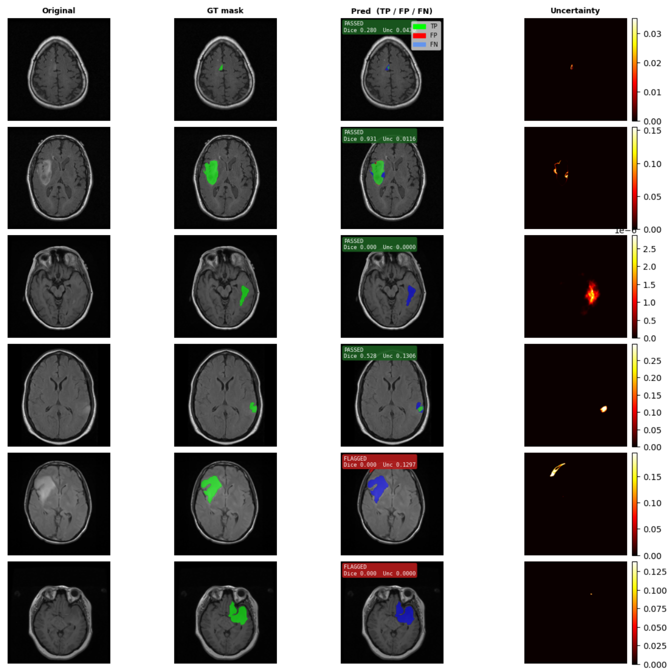


Fig. 4: Uncertainty maps for the brain MRI specialist across evaluation cases. Correctly routed cases show low, boundary-localised uncertainty. Misrouted cases (brain specialist applied to lung X-rays) produce scattered uncertainty hotspots at rib edges and mediastinal borders, anatomically irrelevant regions that the model confuses with tumour boundaries.

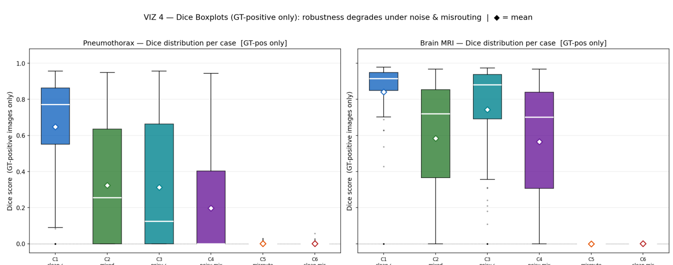


Fig. 5: Dice boxplots per case (GT-positive images only). Specialist models under correct routing achieve tight, high-Dice distributions. Misrouted cases collapse sharply with high variance. The mixed model shows intermediate performance with gradual degradation under noise. Diamond markers indicate case means.

are notably more subtle: under noise, it produces slightly more diffuse boundaries and marginally higher false-positive rates, but the spatial structure of predictions remains anatomically plausible, preventing morphological checks from triggering. This underscores the mixed model's core liability in safety-critical deployment: its errors are structurally plausible and therefore undetectable by post-hoc monitoring.

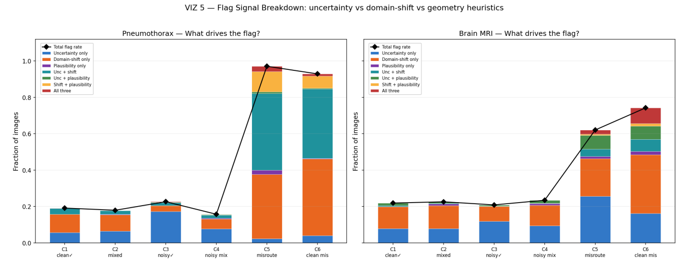


Fig. 6: Flag signal decomposition per case. The Mahalanobis domain-shift signal (orange) is the dominant driver of misroute detection, achieving 100% coverage on pneumothorax misroutes and 75–99% on brain misroutes. Uncertainty (blue) provides complementary coverage. During normal operation (C1–C4), both signals contribute modestly, keeping false-alarm rates low.

Appendix I
Additional Training Techniques

The following techniques are applied during individual model training and standalone model evaluation (i.e. within each organ-specific training notebook), prior to and independent of the failure-aware flagging framework described in the main paper.

Test-Time Augmentation (TTA). During standalone model evaluation, each input image is processed under multiple geometric and scale perturbations, and the resulting logit maps are averaged before applying the sigmoid activation [16]. Specifically, each image is processed at three scales ($s \in \{0.9, 1.0, 1.1\}$), both in its original orientation and horizontally flipped, yielding six forward passes per image. Scaled images are resized and zero-padded to the nearest multiple of 32 to maintain encoder compatibility. The averaged logit map is then thresholded to produce the final binary mask. TTA reduces sensitivity to slight variations in object position and scale at the cost of a $6\times$ increase in inference time. Both the specialist and mixed model training notebooks implement TTA as an optional evaluation mode to measure maximum achievable performance. The failure-aware evaluation results reported in the main paper do not use TTA; all flagging-framework metrics are based on single-pass inference.

Class weight rebalancing. Medical segmentation datasets frequently exhibit severe class imbalance: the majority of samples may contain no positive (lesion) pixels. To prevent the model from learning a trivial all-negative predictor, we employ `WeightedRandomSampler` [17] with inverse class-frequency weighting during training. For each organ domain, the sampling weight for class c is $w_c = 1/(n_c + \epsilon)$, where n_c is the number of training samples in class c (positive vs. negative mask) and $\epsilon = 10^{-6}$. This produces approximately balanced mini-batches (50% positive, 50% negative), ensuring that the model receives sufficient gradient signal from the minority class. In the mixed model, class rebalancing is applied independently per organ domain (separate samplers for lung and brain training loaders) to maintain domain-level balance as well.

Threshold optimisation. The standard decision threshold of 0.5 on the sigmoid output is not necessarily optimal for all models and datasets. During training-notebook evaluation, we perform a post-training grid search over candidate thresholds $\tau \in \{0.25, 0.30, \ldots, 0.60\}$ on the combined validation set, selecting the threshold that maximises the mean per-sample Dice score [18]. For the pneumothorax specialist, the optimal threshold was found to be $\tau^* = 0.60$, reflecting the model's tendency to produce low-confidence predictions on thin pneumothorax regions, where a higher threshold reduces false positives without substantially harming recall. This per-model threshold tuning is a standard part of the individual model development workflow. The failure-aware evaluation framework in the main paper uses a fixed threshold of 0.5 across all models to ensure a controlled, threshold-independent comparison of the flagging mechanism itself.